

Nmap Network Enumeration & Vulnerability Assessment

Metasploitable 2 — Controlled VMware Lab

Target: 192.168.6.139

Scanner: Kali Linux — 192.168.6.128

Network: 192.168.6.0/24

Tool: Nmap 7.99

Academic security assessment performed in an isolated, authorized lab environment.

1. Project Overview

This project demonstrates a structured network enumeration and vulnerability assessment workflow using Nmap against a Metasploitable 2 virtual machine. The assessment was performed inside a controlled VMware lab network.

The workflow covered network configuration, host discovery, port scanning, service/version detection, operating-system detection, and Nmap NSE vulnerability scanning.

2. Lab Environment

Component	Value
Scanner	Kali Linux
Scanner IP	192.168.6.128/24
Target	Metasploitable 2
Target IP	192.168.6.139/24
Network	192.168.6.0/24
Virtualization	VMware
Scanning Tool	Nmap 7.99

3. Network Configuration Evidence

Kali Linux interface evidence shows the scanner address 192.168.6.128/24. The target interface evidence shows Metasploitable 2 at 192.168.6.139/24.

```
Session Actions Edit View Help
zsh: corrupt history file /home/mosab/.zsh_history
(mosab@Mosab)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN gro
up default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel stat
e UP group default qlen 1000
    link/ether 00:0c:29:4e:4d:96 brd ff:ff:ff:ff:ff:ff
    inet 192.168.6.128/24 brd 192.168.6.255 scope global dynamic nopref
ixroute eth0
        valid_lft 1584sec preferred_lft 1584sec
    inet6 fe80::20c:29ff:fe4e:4d96/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue
state DOWN group default
    link/ether ce:49:73:66:49:ab brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
4: br-ce7bce5d812d: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc no
queue state UP group default
    link/ether 8a:ac:bb:eb:be:55 brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-ce7bce5d812d
        valid_lft forever preferred_lft forever
    inet6 fe80::88ac:bbff:feeb:be55/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
5: veth4a39756@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc no
```

```

msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:fa:dd:2a
          inet addr:192.168.6.139  Bcast:192.168.6.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe8a:dd2a/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:73 errors:0 dropped:0 overruns:0 frame:0
          TX packets:68 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:6183 (6.0 KB)  TX bytes:7440 (7.2 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:126 errors:0 dropped:0 overruns:0 frame:0
          TX packets:126 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:35929 (35.0 KB)  TX bytes:35929 (35.0 KB)

msfadmin@metasploitable:~$ _

```

4. Connectivity Test

Command:

```
ping -c 4 192.168.6.139
```

The connectivity test returned 4 packets transmitted, 4 received, with 0% packet loss.

```

(mosab@Mosab)-[~]
$ ping -c 4 192.168.6.139
PING 192.168.6.139 (192.168.6.139) 56(84) bytes of data.
64 bytes from 192.168.6.139: icmp_seq=1 ttl=64 time=0.767 ms
64 bytes from 192.168.6.139: icmp_seq=2 ttl=64 time=7.32 ms
64 bytes from 192.168.6.139: icmp_seq=3 ttl=64 time=1.09 ms
64 bytes from 192.168.6.139: icmp_seq=4 ttl=64 time=0.881 ms

— 192.168.6.139 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 0.767/2.513/7.320/2.777 ms

(mosab@Mosab)-[~]
$ █

```

5. Host Discovery

Command:

```
nmap -sn 192.168.6.0/24
```

The host-discovery scan covered all 256 addresses in the /24 subnet and reported 5 hosts up: 192.168.6.1, 192.168.6.2, 192.168.6.139, 192.168.6.254, and 192.168.6.128.

The target selected for detailed assessment was 192.168.6.139 (Metasploitable 2).

6. Port Scanning

Command:

```
nmap 192.168.6.139
```

The scan identified 23 open TCP ports among the default 1,000 TCP ports scanned. Examples include FTP (21), SSH (22), Telnet (23), SMTP (25), DNS (53), HTTP (80), SMB (139/445), MySQL (3306), PostgreSQL (5432), VNC (5900), IRC (6667), AJP13 (8009), and HTTP/Tomcat (8180).

```
(mosab@Mosab)-[~]
$ nmap 192.168.6.139
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 01:59 -0400
Nmap scan report for 192.168.6.139
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.87 seconds
```

7. Service and Version Detection

Command:

```
nmap -sV 192.168.6.139
```

Version detection identified services including vsFTPD 2.3.4, OpenSSH 4.7p1, ISC BIND 9.4.2, Apache httpd 2.2.8, Samba 3.X-4.X, MySQL 5.0.51a, PostgreSQL 8.3.x, VNC 3.3, UnrealIRCd, Apache Jserv/AJP13, and Apache Tomcat/Coyote on port 8180.

```
(mosab@Mosab)-[~]
$ nmap -sV 192.168.6.139
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 02:05 -0400
Nmap scan report for 192.168.6.139
Host is up (0.0025s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rshcd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi        GNU Classpath grmiregistry
1524/tcp  open  bindshell       Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql      PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.62 seconds
```

8. Operating System Detection

Command:

```
nmap -O 192.168.6.139
```

Nmap identified the target as a general-purpose Linux system and reported Linux kernel 2.6.x, with OS details in the Linux 2.6.9-2.6.33 range.

```
(mosab@Mosab)-[~]
$ nmap -O 192.168.6.139
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 03:33 -0400
Nmap scan report for 192.168.6.139
Host is up (0.00079s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
```

9. Vulnerability Assessment

Command:

```
nmap --script vuln 192.168.6.139
```

The Nmap NSE vulnerability scan identified multiple findings. The most significant evidence included a vsFTPD 2.3.4 backdoor on port 21 with CVE-2011-2523 and a reported root-level result (uid=0(root) gid=0(root)). The scan also reported weak TLS/Diffie-Hellman configurations, SSL POODLE, Slowloris, an RMI registry remote class-loading issue, PostgreSQL TLS weaknesses, an UnrealIRCd backdoor indication, and missing HttpOnly attributes on JSESSIONID cookies.

```
(mosab@Mosab) [~/nmap-network-enumeration-lab]
$ nmap --script vuln 192.168.6.139
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 03:50 -0400
Nmap scan report for 192.168.6.139
Host is up (0.0017s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
| VULNERABLE:
| vsFTPD version 2.3.4 backdoor
| State: VULNERABLE (Exploitable)
| IDs: BID:48539 CVE:CVE-2011-2523
| vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
| Disclosure date: 2011-07-03
| Exploit results:
| Shell command: id
| Results: uid=0(root) gid=0(root)
| References:
| https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
| https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
| https://www.securityfocus.com/bid/48539
| http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
| ssl-dh-params:
| VULNERABLE:
| Anonymous Diffie-Hellman Key Exchange MitM Vulnerability
| State: VULNERABLE
| Transport Layer Security (TLS) services that use anonymous
| Diffie-Hellman key exchange only provide protection against passive
| eavesdropping, and are vulnerable to active man-in-the-middle attacks
| which could completely compromise the confidentiality and integrity
| of any data exchanged over the resulting session.
| Check results:
| ANONYMOUS DH GROUP 1
| Cipher Suite: TLS_DH_anon_WITH_AES_256_CBC_SHA
```

10. Security Findings Summary

#	Finding	Port	Reference / State
1	vsFTPD 2.3.4 Backdoor	21/tcp	CVE-2011-2523 — VULNERABLE (Exploitable)
2	Anonymous DH / Logjam	25/tcp	CVE-2015-4000 — VULNERABLE
3	SSL POODLE	25/tcp	CVE-2014-3566 — VULNERABLE
4	Slowloris	80/tcp	CVE-2007-6750 — LIKELY VULNERABLE
5	RMI Registry Remote Code Execution	1099/tcp	VULNERABLE
6	PostgreSQL SSL/TLS Issues	5432/tcp	POODLE / weak DH / CCS Injection
7	UnrealIRCd Backdoor	6667/tcp	Trojaned version indication
8	HTTP Session Cookie Security	8180/tcp	JSESSIONID without HttpOnly

11. Remediation Recommendations

- Remove or replace vulnerable legacy services and versions, especially vsFTPD 2.3.4 and the indicated backdoored UnrealIRCd build.
- Disable unnecessary exposed services and restrict administrative interfaces to trusted hosts or management networks.
- Disable SSL 3.0, anonymous/export-grade Diffie-Hellman, and weak cryptographic parameters; use current TLS configurations.
- Review and harden the RMI registry configuration to prevent unauthorized remote class loading.
- Update the web server and apply appropriate connection limits, timeouts, and security controls for denial-of-service resistance.
- Configure web session cookies with appropriate HttpOnly, Secure, and SameSite attributes where applicable.
- Remove services that are not required and apply supported software updates across the system.

12. Conclusion

The assessment successfully demonstrated a complete Nmap-based enumeration workflow against a controlled Metasploitable 2 target. The process progressed from network discovery and connectivity validation to port enumeration, service/version identification, operating-system detection, and NSE vulnerability assessment.

The resulting evidence and raw scan outputs are organized in the GitHub project under the scans/, screenshots/, findings/, and report/ directories. The vulnerability findings are documented separately with evidence, impact, and remediation guidance.

This assessment was performed in a controlled lab environment for educational and security-testing purposes.